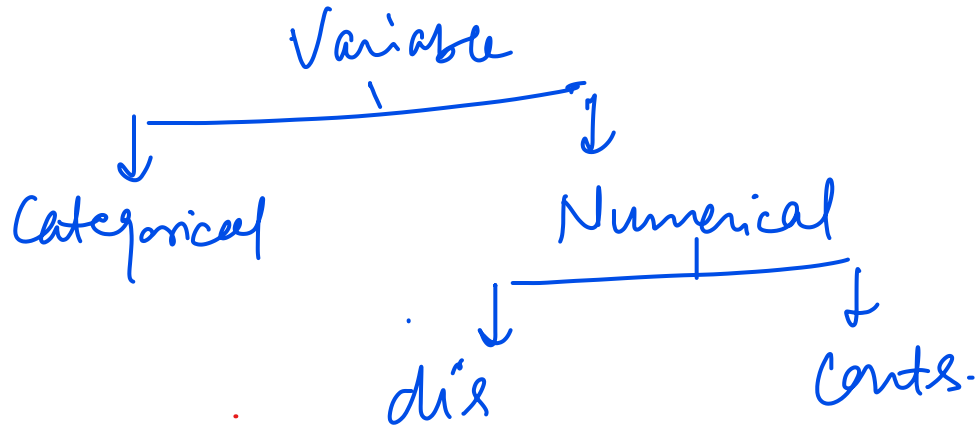


Week - 3 and 4 Revision Session

1.



count.

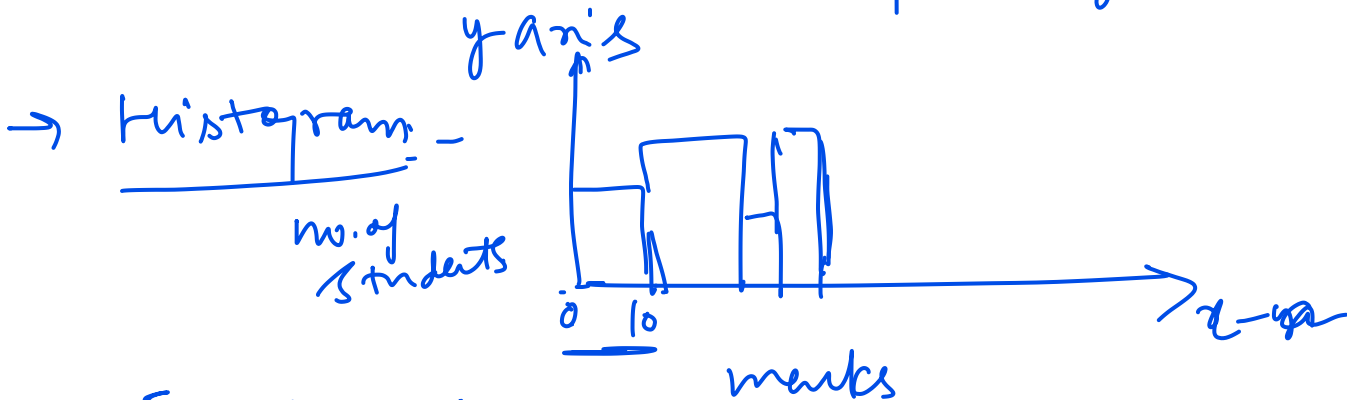
eg: no. of students,



~~measurement.~~

can't count

eg: height, weight.



$[a, b)$
↓ ↓

$[c, d)$

class width = $c - a$.

$[0, 10)$
↓

$[10, 20)$
↓

$$= 10 - 0 = 10$$

* stem & leaf plot

2.

stem	leaf
2	3 4 6
4	3 9
5	0 1 2
6	1 8 9

marks of 11 students

key: 2 | 3 → 23 ✓

23, 24, 26, 43, 49, 50, 51, 52, 61, 68, 69.

Measures of Central tendency.

↳ Mean: $x_1, x_2, \dots, x_n$

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

$$\bar{x} = \frac{1+2+3+4+5}{5} = 3$$

1, 1, 2, 3, 3, 4, 4, 5, ...

x_i f_i $\rightarrow$ $f_i x_i$

1 $\rightarrow$ 2 ✓

2 $\rightarrow$ 1 ✓

3 $\rightarrow$ 2 ✓

4 $\rightarrow$ 2 ✓

5 $\rightarrow$ 1

8

$$\bar{x} = \frac{1+1+2+3+3+4+4+5}{8}$$

$$= \frac{(1 \times 2) + (2 \times 1) + (3 \times 2) + (4 \times 2) + (5 \times 1)}{8}$$

$$\bar{x} = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i} = N$$

Median $\rightarrow$ (1) Arrange the data in ascending order.

(2) n $\begin{cases} \text{even} \\ \text{odd} \end{cases}$

$n \rightarrow \text{even} \therefore \rightarrow$ avg. of $\left(\frac{n}{2}\right)^{\text{th}}$ & $\left(\frac{n}{2} + 1\right)^{\text{th}}$ obsⁿ

$n \rightarrow \text{odd} \therefore$ — median = $\left(\frac{n+1}{2}\right)^{\text{th}}$ obsⁿ

1, 3, 2, 5, 4 $\rightarrow$ 1, 2, 3, 4, 5, 6

4.

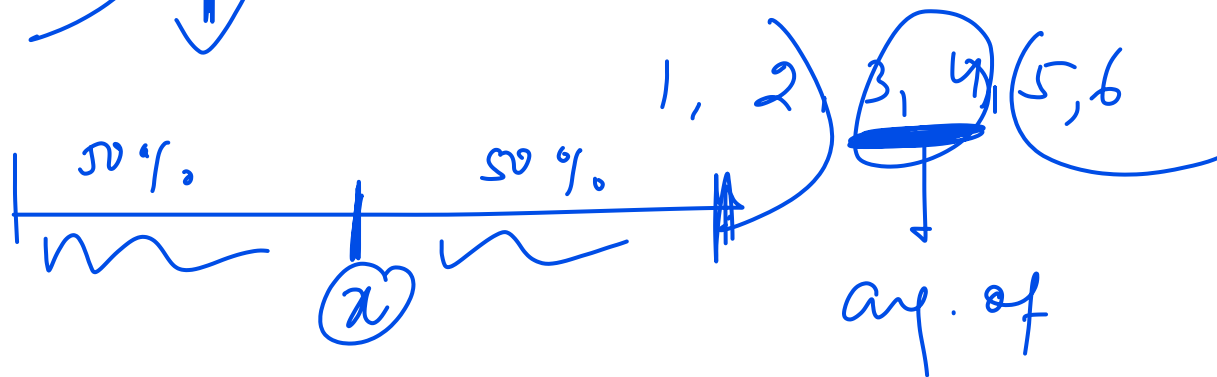
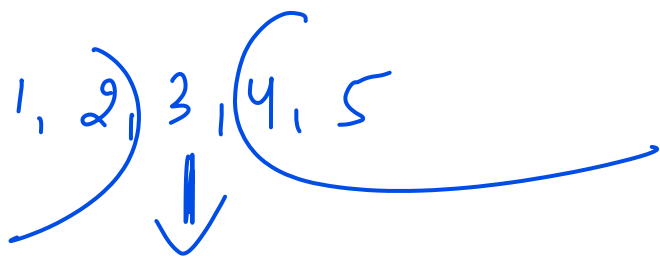
$$\boxed{n=5} \quad \text{md} = \left(\frac{5+1}{2} \right)^{\text{th}} = \frac{6}{2} = 3^{\text{rd}}$$

$$\boxed{\text{md}=3}$$

$$\boxed{n=6} = \text{avg of } \left(\frac{6}{2} \right)^{\text{th}} \text{ \& } \left(\frac{6}{2} + 1 \right)^{\text{th}}$$

$$= \text{avg of } 3^{\text{rd}} \text{ \& } 4^{\text{th}}$$

$$\frac{3+4}{2} = \boxed{3.5}$$



mode:- occurs most frequently.

1, 2, 2, 3, 4, 5, 3

mode = 2, 3 $\rightarrow$ bi-modal.

→ 2, 2, 3, 4, 5

2, 2, 3, 4, 500

$$\text{mean} = \frac{16}{5} = 3.2$$

$$\text{mean} = 102.2$$

$$\text{md} = 3$$

$$\text{md} = 3$$

$$\text{mode} = 2$$

$$\text{mode} = 2$$

2, 2, 3, 4, 5

→ 9, 9, 10, 11, 12

$$\text{mean} = 3.2$$

$$\text{mean} = 10.2 = 3.2 + 7$$

$$\text{md} = 3$$

$$\text{md} = 10 = 3 + 7$$

$$\text{mode} = 2$$

$$\text{mode} = 9 = 2 + 7$$

$x_1, x_2, \dots, x_n$

① → $y_i = x_i + c$

$$\text{new mean} = \text{old mean} + c$$

$$\text{new md} = \text{old md} + c$$

multiply a 'c'

$$y_i = x_i \times c$$

$$\text{new mean} = \text{old mean} \times c$$

$$\text{new md} = \text{old md} \times c$$

$$\text{new mode} = \text{old mode} + c \quad | \quad \text{new mode} = \text{old mode} + c$$

6.

$$\textcircled{1} \text{ Range} = \text{max} - \text{min}$$

$$5 - 1 = \Rightarrow \textcircled{4}$$

1, 2, 3, 4, 500

$$\text{Range} = 500 - 1 = 499 \rightarrow \text{sensitive to outliers.}$$

Variance -

'X'

'Y'

avg: 10, 20, 30, 40, 50 = 30 laphs

avg: 30, 30, 30, 30, 30 = 30 laphs

popⁿ variance: $\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$

Samⁿ var: $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$

6

$$x_i \quad (x_i - \bar{x}) \quad (x_i - \bar{x})^2$$

1 ✓	-2	4
2 ✓	-1	1
3 ✓	0	0
4 ✓	1	1
5 ✓	2	4
$\bar{x} = 3$		10

$$s^2 = \frac{1}{4} \times 10$$

$$\sigma^2 = \frac{1}{5} \times 10$$

'c'

$$\text{new sd} = \text{old sd}$$

new variance = old variance.

multiply by a constant

$$\text{new sd} = c \times \text{old s.d.}$$

$$\text{new var} = c^2 \times \text{old variance}$$

$$\text{s.d. dev.} = Y = 2X + 3$$

$$\text{Var}(Y) = \text{Var}(2X + 3)$$

$$= 4 \times \text{Var}(X)$$

$$\text{Std. dev} \equiv \sqrt{\text{Var}}$$

8.

$$= \sqrt{16 \text{ yr}^2 \text{ var}}$$

$$= 4 \text{ yr}$$

$$\begin{aligned} & \text{yr} \text{ of } \sqrt{4} \\ & x \rightarrow \text{Age} \quad (x_i - \bar{x})^2 \\ & \text{Var} = 16 \text{ yr}^2 \end{aligned}$$

$$\begin{aligned} \text{sd} &= \sqrt{16 \text{ yr}^2} \\ &= 4 \text{ yr} \end{aligned}$$

$$\text{sam. sd} = \sqrt{\frac{1}{n-1} \sum (x_i - \bar{x})^2}$$

$$\text{pop sd} = \sqrt{\frac{1}{n} \sum (x_i - \bar{x})^2}$$

$$\text{sd} = \sqrt{\text{var}}$$

add 'a constant c'

$$\text{new var} = \text{old variance}$$

$$\text{new s.d.} = \text{old s.d.}$$

multiply by a 'c'

$$\text{new var} = c^2 \times \text{old var}$$

$$X \rightarrow \text{Age} \rightarrow \text{yr.}$$

$$\text{new s.d.} = c \times \text{old s.d.}$$

8

$$\text{var} = \frac{1}{n-1} \sum (m_i)^2 = 16 \text{ yr}^2$$

$$s = \sqrt{\frac{1}{n-1} \sum (m_i)^2} = \sqrt{16 \text{ yr}^2} = 4 \text{ yr}$$

Percentiles.

$$s^2 = \frac{1}{n} \sum (x_i - \bar{x})^2 = \frac{1}{n} \left(\sum x_i^2 - n \bar{x}^2 \right)$$

$$s^2 = \frac{1}{n} \sum_{i=1}^n (x_i^2 + \bar{x}^2 - 2x_i \bar{x})$$

$$= \frac{1}{n} \left[\sum_{i=1}^n x_i^2 + \sum_{i=1}^n \bar{x}^2 - 2\bar{x} \left(\sum_{i=1}^n x_i \right) \right]$$

$\bar{x} = \frac{1}{n} \sum x_i$
 $n\bar{x} = \sum x_i$

$$= \frac{1}{n} \left[\sum_{i=1}^n x_i^2 + n\bar{x}^2 - 2\bar{x} \cdot n\bar{x} \right]$$

$$10. = \frac{1}{n} [\sum x_i^2 + \underline{n\bar{x}^2} - \underline{2n\bar{x}^2}]$$

$$= \frac{1}{n} [\sum x_i^2 - n\bar{x}^2]$$

$$s^2 = \frac{1}{n} \sum x_i^2 - \bar{x}^2$$

$$\sum_{i=1}^n x_i + \sum_{i=1}^n x_i$$

Percentiles

25%

25th percentile 75%

30% 30th per. 70%

JEE, GRE

90%

90th per

10%

'Aman'

90%

<

71

Computing percentiles

(1) Arrange the data in ascending order

(2) n p

np — integer — $\uparrow$ avg. of at positions n_p^{th} & $(n_p+1)^{th}$

decimal (not an integer) $\uparrow$ n_p^{th} & $(n_p+1)^{th}$

→ determine the smallest integer greater than np , value at that position is

→ 11, 13, 12, 15, 14
→ 11, 12, 13, 14, 15 ~~2, 3, 4, 5~~

$$n = 5, \quad p = 25\% = 0.25$$

$$\underline{np} = 5 \times 0.25 = 1.25 = 2^{nd} \text{ obs}$$

12

12.

70th percentile

$$n = 5$$

$$p = 0.7$$

$$np = 5 \times 0.7 = 3.5 \rightarrow 4^{\text{th}} \text{ obs'n}$$

11, 12, 13, 14, 15, 16

$$n = 6$$

$$p = 0.3$$

$$np = 6 \times 0.3 = 1.8 \rightarrow 2^{\text{nd}}$$

30th per = 12.

50th percentile

$$p = 0.5$$

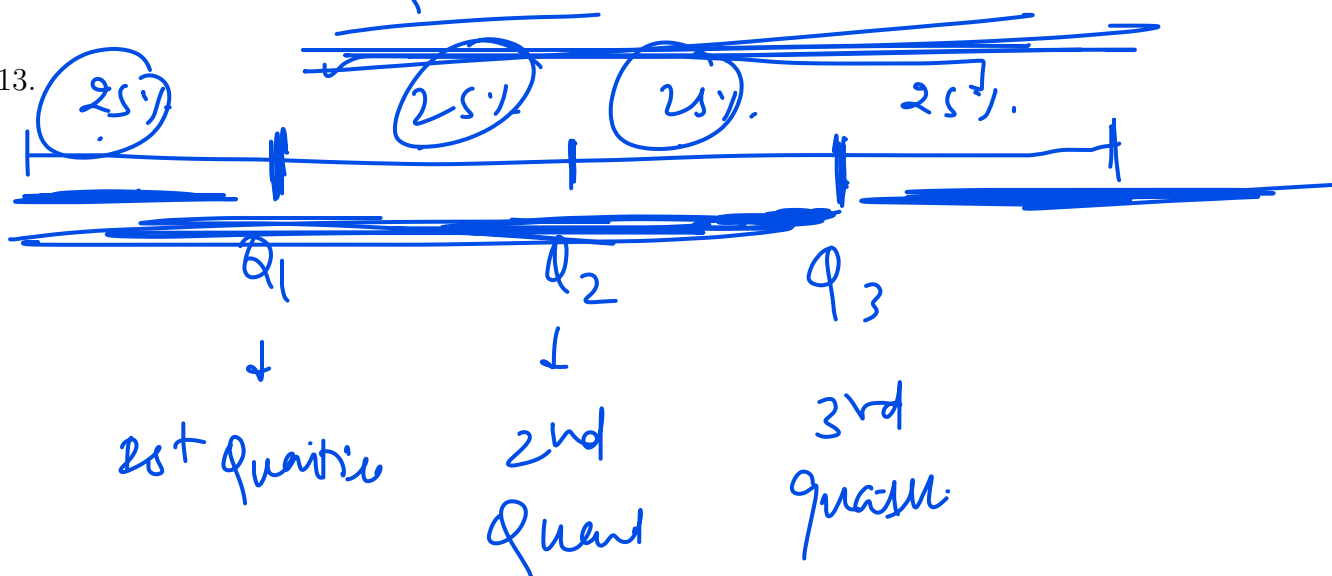
$$np = 6 \times 0.5 = 3$$

avg. of 3rd & 4th obs'n

$$\text{50th pr} = \frac{13 + 14}{2}$$

Quantiles.

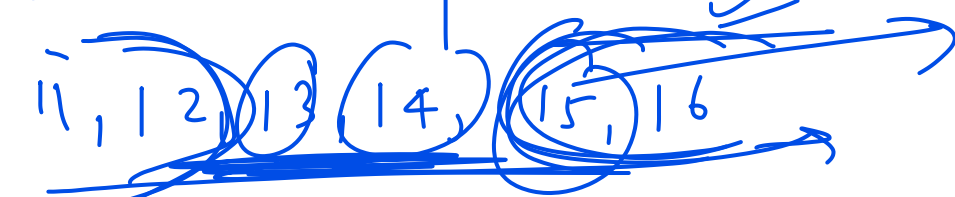
13.



$Q_1 \rightarrow$ 1st quantile $\underline{\quad}$

$Q_2 \rightarrow$ 50th percentile $\underline{\quad}$ median

$Q_3 \rightarrow$ 75th percentile



$$n=6, \quad p=0.25 \Rightarrow 6 \times 0.25 = 1.5 = 2^{\text{nd}} \text{ quantile}$$

$$Q_1 = 12$$

$$n=6, \quad p=0.5 \Rightarrow np = 6 \times 0.5 = 3, \text{ avg of 3rd}$$

$$Q_2 = \frac{13+14}{2}$$

$$n=6, p=0.75 \Rightarrow n_p = 6 \times 0.75 = 4.5 \approx 5^{\text{th}} \text{ obs}^n$$

14.

$$Q_3 = 15$$

Note

$$IQR = Q_3 - Q_1 = 15 - 12 = 3$$

$$11, 12, 13, 14, 15, 16 \approx$$

$$\text{outlier} \leftarrow Q_1 - 1.5(IQR) = 12 - 1.5 \times 3 = 7.5$$

$$Q_3 + 1.5(IQR) \leftarrow \text{outlier} \quad \leftarrow 7.5$$

$$15 + 1.5 \times 3 = 19.5 \quad \text{no outlier}$$

Association

15. ① Association b/w two categorical variables

↳ ① Row Relative freq. ✓

② Colⁿ Rel. Freq. ✓

Row rel.

$$\text{Rel freq} = \frac{f_{ij}}{N}$$

Gender	Own a phone		Row total
	Yes	No	
Female	34/44	10/44	44
Male	42/56	14/56	56
	76/100	24/100	100

f_{ij}/N

Col Rel fr

F 77.27% 22.73%

M 75% 25%

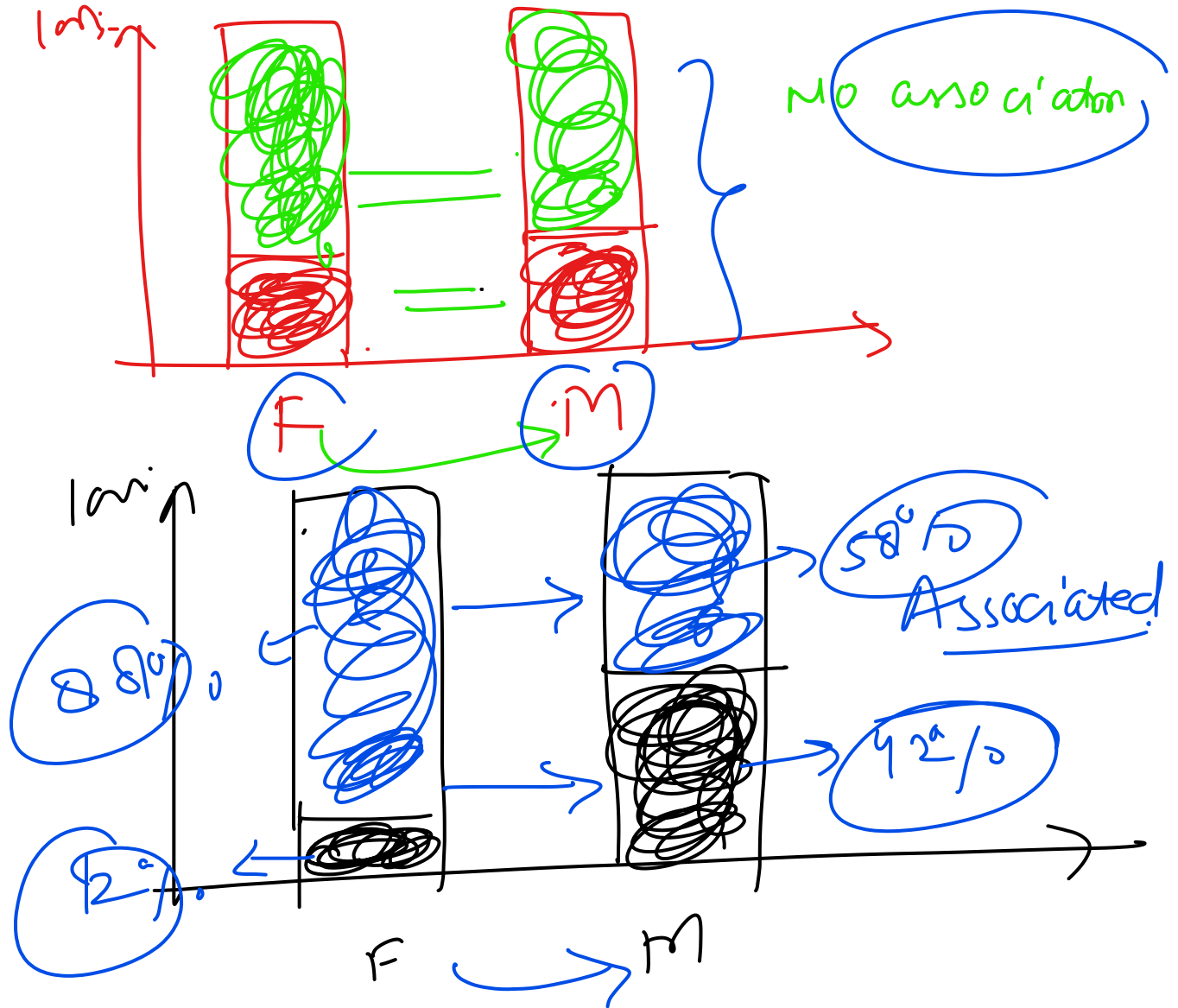
76% 24%

0.76

If row ed. freq are same $\rightarrow$ no association

16.

If row rel. freq are sig. diff $\Rightarrow$ association



Association b/w two numerical var

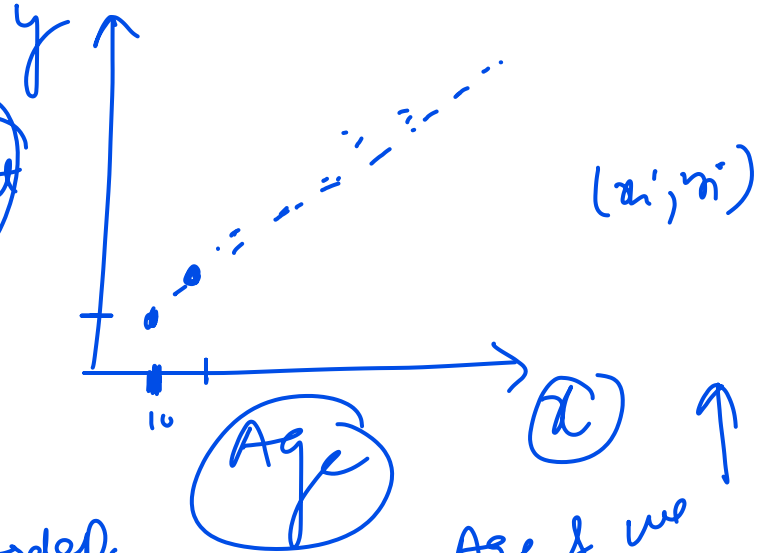
•

scatter plot
 $X \uparrow, Y \uparrow$

indep.
 dep.

(x_i, y_i)
 $\downarrow$

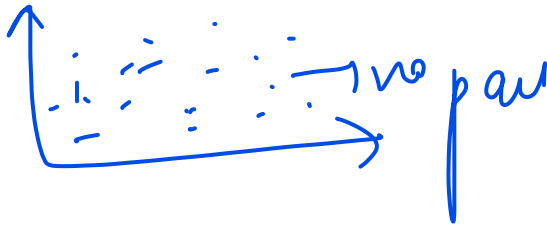
weight



Age indep

weight dep.

Age & we
 X & Y
 $\checkmark$
 are
 positive
 related



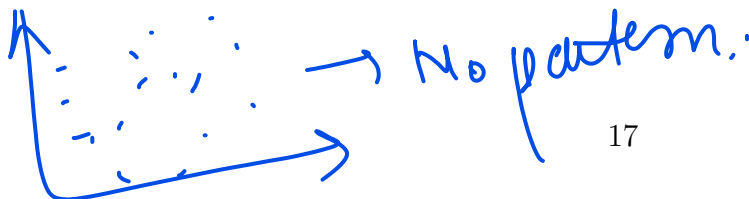
reverse negative association

$X \uparrow, Y \downarrow$

$X \uparrow, Y \uparrow$

$X \downarrow, Y \uparrow$

$X \downarrow, Y \downarrow$



Covariance :- linear association b/w two numerical variables

$$\text{pop}^n \text{ cov} = \frac{1}{n} \sum_{i=1}^n (\underbrace{x_i - \bar{x}}_{\text{+}}) \cdot (\underbrace{y_i - \bar{y}}_{\text{-}})$$

$$\text{sample cov} = \frac{1}{n-1} \sum_{i=1}^n (\underbrace{x_i - \bar{x}}_{\text{+}}) (\underbrace{y_i - \bar{y}}_{\text{-}})$$

$$= \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x}) (x_i - \bar{x})$$

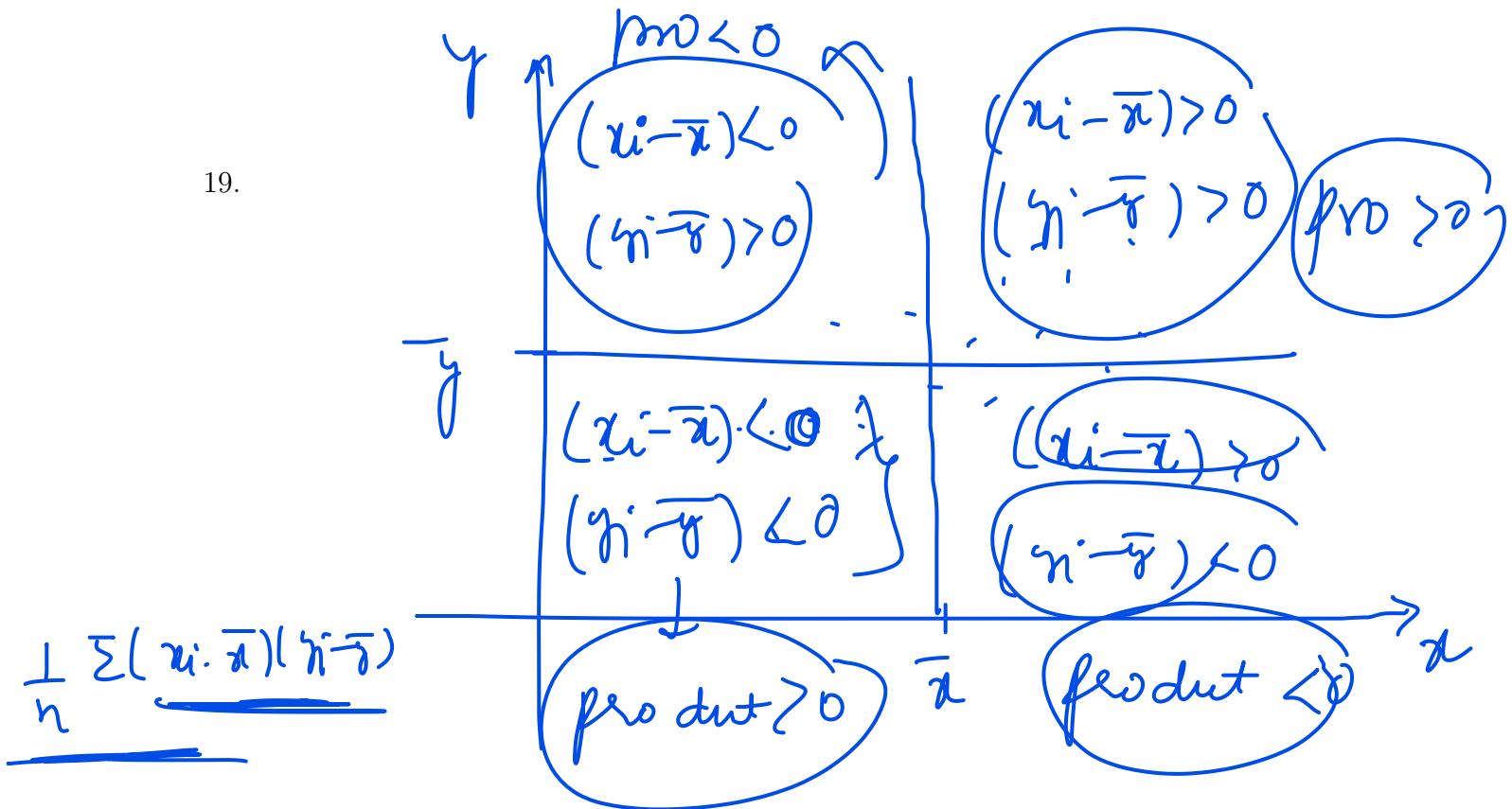
$$= \frac{1}{n} \sum_{i=1}^n \underline{(x_i - \bar{x})^2} = \text{pop}^n \text{ var } x, 0$$

$(\text{Cov}(x, y) = 0 \Rightarrow \text{No linear association b/w } x \text{ \& } y)$

$x : , \text{ } \underline{y = x^2} ;$

$\text{Cov}(x, y) = 0$

19.



correlation $\rightarrow$ Age, weight

$$\text{Cov}(x, y) = 520 \checkmark$$

$$\text{Cov}(x, y) = 152 \checkmark$$

$$\text{Cov}(x, y) = -70 \checkmark$$

$$\text{Cov}(x, y) = -170 \checkmark$$

$$r = \frac{\text{Cov}(x, y)}{\text{sd}(x) \text{sd}(y)} = \frac{\text{gr} \cdot \text{kg}}{\text{gr} \cdot \text{kg}}$$

$$r = \frac{\frac{1}{n-1} \sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\frac{1}{n-1} \sum (x_i - \bar{x})^2} \sqrt{\frac{1}{n-1} \sum (y_i - \bar{y})^2}}$$

520
152

$$r = \frac{\sum (x_i - \bar{x})}{\dots}$$

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \sqrt{\sum (y_i - \bar{y})^2}} \quad \left. \vphantom{\sum (x_i - \bar{x})^2} \right\}$$

20.

$$-1 \leq r \leq 1$$

- $0.75 \leq r \leq 1$ → strong & positive linear relationship
- $0.5 \leq r < 0.75$ → moderate & positive " "
- $0.25 \leq r < 0.5$ → weak & positive _____
- $-0.25 < r < 0.25$ → Absence of linear relationship
- $-0.5 < r \leq -0.25$ → weak & negative linear _____
- $-0.75 < r \leq -0.5$ → moderate & negative _____
- $-1 \leq r \leq -0.75$ → strong & neg lin - -

① Assocⁿ b/w 2 cat. var

② Assoⁿ b/w 2 num var

Association b/w a categorical & a numerical var.

two categories

Absolute point Bi-serial corr. coeff.

$$r_{pb} = \frac{|\bar{y}_0 - \bar{y}_1|}{s_y} \cdot \sqrt{p_0 p_1}$$

popⁿ std. dev.

$$p_0 = \frac{n_0}{n}$$

$$p_1 = \frac{n_1}{n}$$

$$(n-1)s^2 = n\sigma^2$$

$$\sigma^2 = \left(\frac{n-1}{n}\right)s^2 \quad \sigma^2 = \left(\frac{n-1}{n}\right)s^2$$

$$\sigma_x = \sqrt{\frac{n-1}{n}} \cdot s$$

22.

$$s_{pb} = \frac{|\bar{y}_0 - \bar{y}_1|}{\sqrt{\frac{n-1}{n} \cdot s_x}} \sqrt{p_0 p_1}$$

$$s_{pb} = \frac{|\bar{y}_0 - \bar{y}_1|}{\sqrt{\frac{n-1}{n} \cdot s_x}} \sqrt{\frac{n_0}{n} \cdot \frac{n_1}{n}}$$

$$s_{pb} = \frac{|\bar{y}_0 - \bar{y}_1|}{s_x} \sqrt{\frac{n_0}{n} \cdot \frac{n_1}{n-1}}$$

Abgeseh.

$$s_{pb} = \frac{(\bar{y}_0 - \bar{y}_1)}{s_x} \sqrt{p_0 p_1} \xrightarrow{\text{weg.}}$$

$$= \frac{(\bar{y}_1 - \bar{y}_0)}{s_x} \sqrt{p_0 p_1}$$

$$p_0 = \frac{n_0}{n}, \quad p_1 = \frac{n_1}{n}$$

$$s_{pb} = \frac{|\bar{y}_0 - \bar{y}_1|}{s_x} \sqrt{p_0 p_1}$$

22

$$\begin{aligned} (\bar{y}_0 - \bar{y}_1) &\rightarrow \\ (\bar{y}_1 - \bar{y}_0) &\end{aligned}$$

/ E

Gender m marks

F	→ 1 ✓	98
M	→ 0 ✓	92
M	→ 0 ✓	70
F	→ 1 ✓	68
M	→ 0 ✓	56
F	→ 1 ✓	80

$$p_0 = \frac{n_0}{n} = \frac{3}{6} = 0.5$$

$$p_1 = \frac{n_1}{n} = \frac{3}{6} = 0.5$$

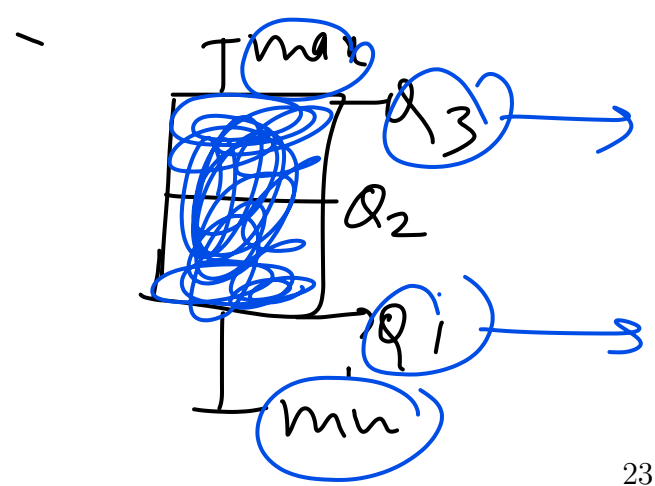
$$\bar{y}_0 = \frac{92 + 70 + 56}{3} = \checkmark$$

$$\bar{y}_1 = \frac{98 + 68 + 80}{3} \checkmark$$

$$z_{pb} = \frac{|\bar{y}_0 - \bar{y}_1|}{\sigma_x} \sqrt{p_0 p_1}$$

pop ⁿ 800

$$z_{pb} =$$



24.

25.

26.

27.

28.

29.

30.

31.

32.

33.

34.

35.